

# Dilan Karaguler

Ph.D. Candidate in Mathematics, Michigan State University

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[Google Scholar](#) | [GitHub](#) | [LinkedIn](#) | [dilankaraguler.github.io](https://dilankaraguler.github.io)

## EDUCATION

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### Michigan State University

Ph.D. in Mathematics (Candidate)

– Advisors: Prof. Ekaterina Rapinchuk and Prof. Guowei Wei

East Lansing, MI, US

August 2022 – May 2027 (Expected)

### Middle East Technical University (METU)

M.S. in Mathematics

– Thesis: *A Survey on Multidimensional Persistence Theory*

– Advisor: Assoc. Prof. Semra Pamuk

Ankara, Türkiye

September 2019 – September 2021

### Middle East Technical University (METU)

B.S. in Mathematics, Honor Student

Ankara, Türkiye

June 2014 – June 2019

## TEACHING LEADERSHIP & MENTORING

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### Lead Teaching Assistant

Department of Mathematics, Michigan State University

August 2024 – Present

East Lansing, MI, US

- **2026 – 2027:** Co-lead, with Eloy Moreno-Nadales, a yearlong workshop series that prepares first-year graduate students for their first teaching assignments. Sessions so far: an introduction run as a gallery walk, with a “Math PhD passport” and department scavenger hunt connecting new students with senior graduate students; and *Three Aspects of Teaching* (clarity and delivery, communication and language, engagement and cohort needs), built on the TAs’ own classroom observations to prepare their mock teaching.
- **Fall 2025 (MTH 299):** Prepared recitation worksheet materials, gathered TA feedback for the instructors, and substituted for recitation leaders when needed.
- **Spring 2025 (MTH 133):** Ran TA meetings, prepared weekly recitation worksheets, and set up Teams tools to track the TA timeline and grading deadlines.
- **Fall 2024:** Ran the orientation for undergraduate learning assistants (ULAs) starting in August, held mock-teaching and feedback sessions, and observed ULAs and GTAs during the semester to give feedback; as MTH 133 pod leader, ran meetings with recitation leaders and prepared weekly recitation worksheets.

### Graduate Lead, Center for Instructional Mentoring (CIM)

Department of Mathematics, Michigan State University

Summer 2024

East Lansing, MI, US

- Mentored first-time graduate instructors through weekly check-ins, one-on-one conversations and non-evaluative teaching feedback, and introduced the teaching technology they could use.
- Created a teaching handbook built on interviews with senior GTAs about the courses they taught, describing each course’s expectations so junior graduate students can choose assignments that fit their needs.

### Mentor, Academic Culture and English Skills (ACES) Program

Michigan State University

Summer 2024

East Lansing, MI, US

- Mentored incoming international graduate students preparing to become MSU teaching assistants, introducing U.S. classroom culture, university structure and cross-cultural communication, and organized social events.

### Supervisor, Math Learning Center (MLC)

Michigan State University (Teaching Exemption Fellowship, 2022 – 2023)

Fall 2022 – Spring 2023, Fall 2024

East Lansing, MI, US

- Managed how tutors were distributed among students and arranged the room to match demand.

**Student feedback:** [dilankaraguler.github.io/teaching/what-students-say](https://dilankaraguler.github.io/teaching/what-students-say)

## COURSES TAUGHT

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| <b>MTH 314 — Matrix Algebra with Computational Applications</b>                                                                                                                                                                             | Fall 2026                |
| Instructor (1 section), Michigan State University                                                                                                                                                                                           | East Lansing, MI, US     |
| – Prepare and deliver lectures, facilitate group work, and hold help-room hours.                                                                                                                                                            |                          |
| <b>MTH 299 — Transitions</b>                                                                                                                                                                                                                | Fall 2025                |
| Teaching Assistant (2 sections), Michigan State University                                                                                                                                                                                  | East Lansing, MI, US     |
| – Co-taught with Ester Sgallova: led weekly recitations, wrote homework solutions, graded homework and quizzes, and held help-room hours.                                                                                                   |                          |
| <b>MTH 314 — Matrix Algebra with Computational Applications</b>                                                                                                                                                                             | Spring 2025              |
| Teaching Assistant (1 section), Michigan State University                                                                                                                                                                                   | East Lansing, MI, US     |
| – Graded homework and the final exam, ran quiz and exam review sessions, and held help-room hours.                                                                                                                                          |                          |
| <b>MTH 133 — Calculus II</b>                                                                                                                                                                                                                | Fall 2024                |
| Teaching Assistant (1 section) and pod leader, Michigan State University                                                                                                                                                                    | East Lansing, MI, US     |
| – Led a recitation section; as pod leader, ran meetings with the recitation leaders and prepared the weekly recitation worksheets.                                                                                                          |                          |
| <b>MTH 235 — Differential Equations</b>                                                                                                                                                                                                     | Spring 2024              |
| Teaching Assistant (3 sections), Michigan State University                                                                                                                                                                                  | East Lansing, MI, US     |
| – Wrote recitation notes with lecture summaries and practice problems, graded quizzes, held help-room hours, and followed up with students in online sections.                                                                              |                          |
| <b>MTH 133 — Calculus II</b>                                                                                                                                                                                                                | Fall 2023                |
| Teaching Assistant (3 sections: 2 online, 1 in person), Michigan State University                                                                                                                                                           | East Lansing, MI, US     |
| – Prepared weekly lecture summaries for the recall session before group work and worksheets; graded and proctored worksheets and quizzes.                                                                                                   |                          |
| – Held help-room hours and tracked student progress, reaching out to students as needed, especially in online sections.                                                                                                                     |                          |
| <b>MTH 133 — Calculus II</b>                                                                                                                                                                                                                | Summer 2023              |
| Teaching Assistant (1 online asynchronous section), Michigan State University                                                                                                                                                               | East Lansing, MI, US     |
| – Contacted students weekly by email and tracked their course activity.                                                                                                                                                                     |                          |
| <b>MTH 154, 153, 120, 119 — Calculus I and II</b>                                                                                                                                                                                           | January 2020 – June 2022 |
| Graduate Teaching Assistant, Middle East Technical University                                                                                                                                                                               | Ankara, Türkiye          |
| – MTH 154 Calculus II for Mathematics Students (Spring 2022), MTH 153 Calculus I for Mathematics Students (Fall 2021), MTH 120 Calculus II for Engineering Students (Spring 2021), MTH 119 Calculus I for Engineering Students (Fall 2020). |                          |
| – Led recitations and office hours, graded, and proctored online and in-person exams.                                                                                                                                                       |                          |
| <b>Calculus for Mathematics Students</b>                                                                                                                                                                                                    | March 2018 – June 2019   |
| Undergraduate Teaching Assistant, Middle East Technical University                                                                                                                                                                          | Ankara, Türkiye          |
| – Graded homework and held review hours.                                                                                                                                                                                                    |                          |

## AWARDS, FELLOWSHIPS & SCHOLARSHIPS

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|-----------------------------------------------------------------------------------------------|-------------|
| Doctoral Student Summer Internship Fellowship, The Graduate School, Michigan State University | Summer 2026 |
| <b>Excellence in Teaching Award</b> , Department of Mathematics, Michigan State University    | May 2025    |
| Colleges Online Learning Academy (COLA) Fellowship, Michigan State University                 | Summer 2024 |
| Teaching Exemption Fellowship, Michigan State University                                      | 2022 – 2023 |
| RONESANS Scholarship, RONESANS                                                                | 2018 – 2019 |
| UNIGENC Scholarship, UNIGENC Association                                                      | 2015 – 2019 |
| Merit Scholarship, Turkish Ministry of National Education                                     | 2010 – 2014 |

## TEACHING DEVELOPMENT & EDUCATIONAL RESEARCH

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### Program Evaluator & Instructional Researcher

May 2026

AI for Advanced Semiconductor Manufacturing Workshop, Michigan State University

East Lansing, MI, US

- Designed pre/post-workshop surveys and observation protocols to evaluate program effectiveness and adult learning outcomes.
- Analyzed participant feedback quantitatively and with qualitative thematic coding, tracking confidence growth across AI and semiconductor manufacturing topics.
- Wrote an evaluation report grounded in adult learning theory with curriculum recommendations for the workshop.

### Colleges Online Learning Academy (COLA) Scholar

Summer 2024

Michigan State University

[Portfolio](#)

- Built a teaching portfolio website and wrote a project plan for online course design as program deliverables.

## PUBLICATIONS

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**D. Karaguler** and G.-W. Wei. *Mayer Path Homology*. Foundations of Data Science, accepted (2026).  
[doi:10.3934/fods.2026025](https://doi.org/10.3934/fods.2026025), [arXiv:2605.16525](https://arxiv.org/abs/2605.16525).

**D. Karaguler**. *A Survey on Multidimensional Persistence Theory*. M.S. Thesis, Middle East Technical University (2021).

**D. Karaguler** and G.-W. Wei. *Persistent Mayer Path Laplacians*. In preparation (2026).

## TALKS

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### Mayer Path Homology

April 2026

TDA Seminar, Dept. of Computational Mathematics, Science and Engineering, MSU

East Lansing, MI, US

### Interaction Topology in TDA

October 2025

Invited lecture, MTH 994: Mathematics-Assisted AI and AI-inspired Mathematics, MSU

East Lansing, MI, US

### Mayer Topology in TDA

October 2025

Invited lecture, MTH 994: Mathematics-Assisted AI and AI-inspired Mathematics, MSU

East Lansing, MI, US

### From Data to Barcodes

April 2021

Graduate Student Seminar, Middle East Technical University

Ankara, Türkiye

## POSTER PRESENTATIONS

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### *Mayer Path Topology and Persistence*

September 2025

SIAM Great Lakes Section Meeting, Illinois Institute of Technology

Chicago, IL, US

### *Mayer Path Topology and Persistence*

March 2025

Mathematics of Data, Dynamics, and Life Sciences Conference, UC Irvine

Irvine, CA, US

## RESEARCH EXPERIENCE

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### Graduate Researcher

August 2022 – Present

Michigan State University

East Lansing, MI, US

- Develop Mayer path homology, a homology theory for directed graphs built from  $N$ -nilpotent differentials on path complexes; published in *Foundations of Data Science* (with G.-W. Wei).
- Implement topological feature extraction (path complexes, Mayer homology and Laplacians, persistent homology) in Python with GUDHI, NumPy and PyTorch, and combine these features with graph neural networks.

### Research Mentor

May 2023 – June 2023

Topology RTG, Michigan State University

East Lansing, MI, US

- Mentored an undergraduate researcher on a project in Morse theory: project design, methodology and research presentation; the student went on to major in mathematics.

## INDUSTRY EXPERIENCE

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### Software Engineer Intern

May 2026 – August 2026

Intel Corporation

Oregon, US

- Built an end-to-end graph transformation pipeline in Python that extracts schematic topology directly from chip layout files, without pre-existing schematic metadata.
- Modeled low-level device instances as graph nodes and extracted a local subgraph around each instance to capture its connectivity.
- Developed graph aggregation and embedding routines producing a vector representation per device, used to classify components as analog, digital or fill devices.
- Worked with Design Yield and Manufacturability engineers to evaluate the prototype for automated structural recognition in layout verification.

### Graduate Technical Intern

May 2025 – August 2025

Intel Corporation

Oregon, US

- Performed spatial pattern analysis on semiconductor chip layer images with KLayout and Python to identify and cluster recurring structures across layout datasets.
- Applied graph neural networks to search for and classify similar patterns across multi-layer designs.
- Partnered with cross-functional yield teams to bring ML-based pattern recognition into automated yield-analysis workflows.

## SERVICE, LEADERSHIP & OUTREACH

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### Graduate Student Representative, Graduate Studies Committee

2025 – 2026

Department of Mathematics, Michigan State University

East Lansing, MI, US

- Represented all mathematics graduate students at committee meetings with faculty.
- Planned and prepared Responsible Conduct of Research (RCR) workshops with the Graduate Chair.
- Met with prospective graduate students and hosted graduate student town hall meetings on the state of the department.
- Served as a point of contact for graduate students with grievances and directed them to department and university resources.

### President / Board Member

August 2022 – August 2025

Association for Women in Mathematics (AWM), MSU Chapter

East Lansing, MI, US

- Organized seminars, career workshops, informational sessions on academic life and social events for women in mathematics.

### Member, First-Generation/Low-Income (FGLI) Group

September 2024 – May 2025

Department of Mathematics, Michigan State University

East Lansing, MI, US

- Took part in community gatherings and social events with first-generation and low-income undergraduates.

### Organizing Committee Member

April 2024

Graduate Student Topology and Geometry Conference, Michigan State University

East Lansing, MI, US

- Handled communication with speakers and attendees, room assignments and talk scheduling, and moderated sessions.

### President

February 2015 – December 2018

METU Math Club

Ankara, Türkiye

- Organized seminars, workshops and regular reading groups in topology, algebra and other areas.
- Led Math Club reading sessions with first-year students in abstract algebra and number theory, preparing each student to lead a discussion of part of the material.

## REFERENCES

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Available upon request.